

The Latest In STORAGE PRODUCTS



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Let us take a look at what a minute looks like to a Big Data analyst.

- 20.8 million messages sent on WhatsApp
- 2.4 million searches generated on Google
- 2.78 million videos watched on YouTube
- 1.04 million videos watched on Vine
- 700,000 people logging on to Facebook

Looks like a lot of big numbers just for social media. Even with this rise of social media, millions of emails are still sent every minute. Despite an estimated international Internet connectivity at about 3.2 billion, those are some very huge numbers. So where does this data go?

Data centres defining bulk storage

Storing all recorded information requires heavy-duty data storage devices. This is usually done in data centres. Once stored, accessing these is another concern, which calls for specialised data storage devices with ultra-high-speed data-access capabilities. These have to be easily accessible with fast and accurate calculations and analytics being done on these.

Onboard, the memories forgotten. Bulk storage of data calls for huge storage spaces. This increases the access time even with the best of systems. Solution to this comes in the form of various onboard memories, ranging from random access memory (RAM) to registers, which are very fast, and hence improve the computational calibre of the system.

We often fail to consider onboard memories while discussing storage. Consumerisation of memory has provided easy access to storage devices like thumb drives and external hard disk drives (HDDs), making registers and embedded memories just for design and development engineers. With data speeds of about 400MB/sec under ideal conditions, these are still far behind embedded dynamic random access memory (DRAM).

Memory design: Experts talk

Arun Kumar, sales and marketing manager, Toshiba India Pvt Ltd, says, "Flash memo-

SPEED COMPARISON FOR ONBOARD MEMORIES

Memory type	Bandwidth
Standard (256k×16)	240MB/sec
High speed (256k×16)	400MB/sec
2MB×8 RDRAM (rambus DRAM)	500MB/sec
SDRAM (synchronous DRAM)	640MB/sec
Embedded DRAM (32k×256)	2560MB/sec

ries have realised an immense range of applications that include memory cards, solid-state drives (SSDs) and even industrial equipment."

Today we have a lot of memory devices ranging from multi-TB hard drives to registers in kilobytes that have to be designed and developed according to the processing capabilities. Vivian Singh, chief executive officer, Strontium Technology India Pvt Ltd, adds, "Storage products need to meet the physical settings and fulfil the performance/behaviour with various hosts."

Any storage can be measured on the basis of three basic factors. First, cost of the memory module must be very low; second, it has to be very fast; and third, it has to be able to store data when power is switched off, that is non-volatile.

Development challenges. Memory is an integral part of any electronic system. Hence, there is a need for memory to be fast and sized according to the application. Additionally, according to Singh, "There is always the challenge to bring the laboratory experiment to practical and actual mass production." On one hand we can create an ideal memory in the lab, but on the other if it raises production challenges, it would not help in the development of storage.

Designing the system and employing memory. Once developed, implementing the memory into devices is another concern. Suitable buses are required for data access that do not compromise data access speeds.